MEDICAL INSURANCE EXPENDITURE ANALYSIS: AN EMPIRICAL DIAGNOSTIC AND MODELLING STUDY

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ABSTRACT

Accurate healthcare cost prediction is vital for insurance risk assessment and policy pricing. This project evaluated predictive modeling techniques using multi-attribute health data (n=1,338) to forecast individual annual medical charges. The exploratory phase revealed severe non-linear relationships, heteroscedasticity and multicollinearity, which were addressed using Variance Inflation Factor (VIF) diagnostics. Power transformations and polynomial interaction terms were engineered to capture non-linear interactions across demographic and lifestyle attributes. Multiple Linear Regression (OLS) and non-linear Random Forest Regressors were trained using 5-fold cross-validation. Evaluating model performance on a holdout test set demonstrated that feature engineering combined with Random Forest modeling outperformed the baseline OLS model, increasing the test R^2 from 0.784 to 0.863 and reducing the Mean Absolute Error (MAE) by 37.2%

INTRODUCTION AND PROBLEM FORMULATION

The models associated with healthcare pricing depends accurately on quantifying risk based on individual demographic and health characteristics. Linear estimation methods often fail to capture complex non-linear behaviors, such as the compounding interaction between age, body mass index (BMI) and smoking habits.

This project addresses three primary research objectives:

- Identifying key non-linear drivers and interactions influencing individual medical expenditures.
- Mitigating multicollinearity using Variance Inflation Factor (VIF) diagnostics to stabilize parameter estimations.
- Quantifying the performance again achieved by combining polynomial feature transformations with non-linear ensemble models over standard Ordinary Least Squares (OLS) regression.

DATASET EXPLORATION & STRUCTURAL DIAGNOSTICS

The analysis utilized a dataset of health insurance containing 1,338 individual records which is being evaluated across 7 attributes which are as follows:

- Age: Continuous demographic attribute ranging from 18 to 64 years.
- Sex: Categorical Binary indicator (female, male).
- BMI: Continuous body mass index metric (Kg/m^2).
- Children: Discrete count of covered dependents.
- Smoker: Categorical indicator of tobacco use status (yes, no).

- Region: Categorical geographic classification (northeast, northwest, southeast, southwest).
- Charges: Continuous dependent variable representing annual medical expenditures.

Key diagnostic findings from the exploratory analysis:

- Bimodal Distribution: The target variable charges displayed heavy right-skewness (mean=$ 13,270.42, median=$9,382.03), driven by high-cost medical claims.
- Primary Cost Driver: Tobacco usage emerged as the strongest predictor of elevated expenditures. Average annual charges of non-smokers are $8,434.27, whereas that of smokers are $32,050.23.
- Interaction effects: A clear non-linear relationship was observed between BMI and smoker status. For non-smokers, increased BMI resulted in increase of marginal charge; for smokers with BMI>=30, charges increased rapidly, demonstrating a strong multiplicative effect.

PREPROCESSING AND FEATURE ENGINEERING PIPELINE

To prepare the dataset for predictive modelling:
1. Categorical Encoding: One-hot encoding converted sex, smoker, and region into numeric binary vectors, dropping one category per attribute to avoid the dummy variable trap.

2.Multicollinearity Removal (VIF): Variance Inflation Factor diagnostics were evaluated across all predictors. Features exhibiting high collinearity (VIF>5) were transformed or dropped to stabilize coefficient estimation.
3. Polynomial Feature Generation: Higher-order polynomial terms (age^2,bmi^2) and explicit interaction terms (BMI x smoker_yes , age x smoker_yes) were engineered to model non- linearity in boundaries within linear frameworks.
4. Feature Standardization: Z-score standardization was applied across continuous features to ensure uniform scaling across distance – sensitive algorithms:
X_scaled= (X – mean)/ std_dev

MODEL TRAINING AND EXPERIMENTAL RESULTS

The dataset was partitioned into an 80% training set and a 20% holdout test set. To ensure generalization, 5-fold cross- validation was implemented across all model training iterations.
Model Evaluation Metric Comparison:

- **Baseline Multiple Linear Regression (OLS):**
  - Test R^2: 0.784.
  - Mean Absolute Error (MAE): Baseline.
- **Engineered Random Forest Regressor (5-Fold CV):**
  - Test R^2: 0.863(+10.1% Relative Improvement).
  - Mean Absolute Error (MAE): -37.2% Reduction vs Baseline.

Key Analytical Observations:

- OLS Performance Limitations: The Multiple Linear Regression model captured general linear trends (R^2=0784), but residual plots displayed heteroscedasticity, underestimating high-cost claim instances.
- Ensemble Model Advantage: The Random Forest Regressor effectively captured complex non-linear feature space splits and higher-order interactions, achieving a test R^2 of 0.863 while reducing MAE by 37.2% compared to the baseline model.
- Feature Importance Analysis: Model interpretation metrics confirmed that smoker_yes, the BMI x smoker_yes interaction term, and age accounted for over 80% of total predictive power.

CONCLUSION

- Non- linear interactions particularly the compound effect of elevated BMI and smoking drive the majority of high-cost insurance claims.
- Incorporating polynomial interaction terms alongside non- linear ensemble techniques substantially improves predictive accuracy over traditional linear approaches.

Future Extensions: Future work will incorporate hyperparameter optimization via Bayesian Search, evaluate Gradient Boosting frameworks (such as XGBoost and LightGBM), and apply SHAP (Shapley Additive exPlanations) values to enhance individual prediction interpretability for underwriting applications.